

VENKATA NAGA POOJA PRATHIPATI

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PROFESSIONAL SUMMARY

AI/ML Engineer with 4+ years of professional experience designing and deploying scalable machine learning, NLP, and generative AI solutions across finance, banking, and enterprise domains. Skilled in building end-to-end AI/ML pipelines, integrating custom models into cloud production systems, and applying responsible AI best practices. Strong expertise in Python, ML frameworks (PyTorch, TensorFlow), and cloud platforms (AWS, Azure, GCP). Published researcher in federated learning and privacy-preserving AI. Seeking to contribute innovative, production-ready AI systems that align with organizational strategies.

CORE SKILLS

- **Programming & Data:** Python, R, SQL, Java, Scala, Pandas, NumPy, Apache Spark, Hadoop
- **AI/ML:** Supervised, Unsupervised, Deep Learning, NLP, Transformers (BERT, GPT, LLaMA, Mistral), Generative AI, RAG, Multi-modal AI
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, LangChain, CrewAI
- **Pipelines & MLOps:** MLflow, Kubeflow, Docker, Kubernetes, Jenkins, CI/CD, Model Monitoring
- **Cloud Platforms:** AWS (SageMaker, Lambda, EMR, EC2, S3), Azure ML Studio, GCP AI Platform, BigQuery
- **Databases:** PostgreSQL, MySQL, MongoDB, Cassandra, Redis, Snowflake
- **Visualization:** Tableau, Power BI, Matplotlib, Seaborn, Plotly
- **Responsible AI:** Bias mitigation, hallucination filters, security guardrails, privacy-preserving ML

PROFESSIONAL EXPERIENCE

Morgan Stanley – New York, NY

Senior AI/Machine Learning Engineer | Jan 2024 – Present

- Supported the development and deployment of advanced AI solutions, including fine-tuned LLMs (GPT-J, LLaMA) using LoRA, QLoRA, RLHF, improving document understanding accuracy by 40%.
- Architected **end-to-end AI pipelines** (data ingestion, preprocessing, model training, evaluation, and deployment) using FAISS, LangChain, Ray Serve, and integrated them into production systems on AWS SageMaker & Kubernetes.
- Built **retrieval-augmented generation (RAG) systems** handling 500K+ documents, enabling real-time semantic search for traders.
- Delivered AI solutions aligned with **enterprise AI strategy**, improving decision support workflows by 35%.
- Implemented **responsible AI frameworks** with content filtering, bias mitigation, and hallucination reduction protocols.
- Collaborated with data scientists, product managers, and engineers to **translate business needs into scalable AI systems**.

Institute for Development and Research in Banking Technology (IDRBT) – India

AI/ML Engineer | May 2020 – Apr 2023

- Designed and deployed **ML models for AML detection**, reducing manual review by 45%.
- Implemented **chatbots and NLP systems** automating 80% of customer service requests.
- Built predictive **loan default risk models** (ensemble methods), improving risk accuracy by 33%.
- Developed **real-time fraud detection** systems using anomaly detection and behavioral analytics (92% detection accuracy).
- Created **document automation solutions** with computer vision + OCR, cutting KYC processing time by 67%.
- Built customer segmentation and recommendation systems, boosting cross-selling revenue by 29%.
- Integrated models into **Azure ML Studio** pipelines with automated ETL and regulatory reporting workflows.

EDUCATION

- Master of Science in Computer Science, University of Central Missouri, USA
- Bachelor of Technology in Computer Engineering, IIITDM Kurnool India

PUBLICATIONS

- Federated Scaling of Pre-trained Models for Deep Facial Expression Recognition – Springer, 2024